

Deep Learning

Lecture 1: Introduction

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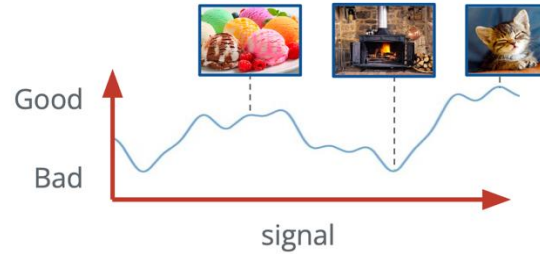


Lecture Overview



1 Introduction

- Definitions
- Examples



2 Learning in Nature

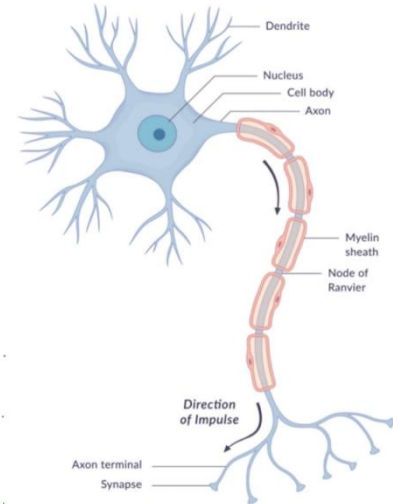
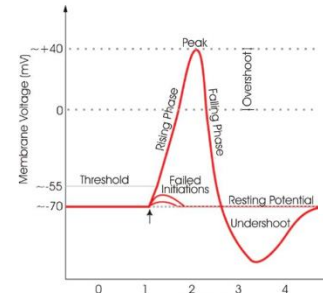
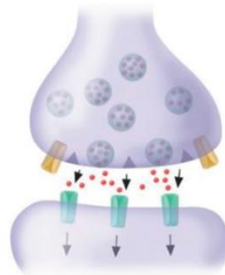
- What is learning?
- How does the brain work?

- Synaptic Plasticity
- Hebbian Theory

3 A Brief History

4 Key Concepts

- Machine Learning
- Reinforcement Learning
- Three Spaces





The aims of the course are:

1. To be able to solve complex ill-defined problems that require deep layers of learning.
2. To understand natural learning, and how it connects to the theory.
3. To ask the right scientific questions given a new task, and use modern deep learning libraries to effectively design, train & test.



Learning in Nature

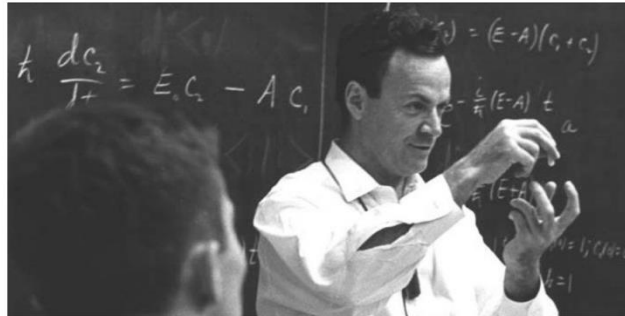
Definition of Learning



Definition:

“We define learning as the transformative process of taking in information that—when internalized and mixed with what we have experienced—changes what we know and builds on what we do. It’s based on input, process, and reflection. It is what changes us.”

Tony Bingham and Marcia Conner





Definition:

“Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. These methods have dramatically improved the state-of-the-art in speech recognition, visual object recognition, object detection and many other domains such as drug discovery and genomics. Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm.”

Yann LeCun, Yoshua Bengio & Geoffrey Hinton
Deep Learning, 2015, Nature - [Read Online](#)

[On the Number of Linear Regions of Deep Neural Networks \(Montúfar et al.\)](#)

[The Power of Depth for Feedforward Neural Networks \(Eldan et al.\)](#)



Examples: Images and Vision

Non-photorealistic interpolation:

- Photos - [video](#)
- Paintings - [video](#)
- DeepFakes - [video](#)

End-to-end self-driving:

- Wayve - [video](#)

Examples: Multimodality

Examples from text:

- GPT-4 and **multimodality**
 - **Input:** The internet
 - **Output:** Any multimodal task

Examples with audio:

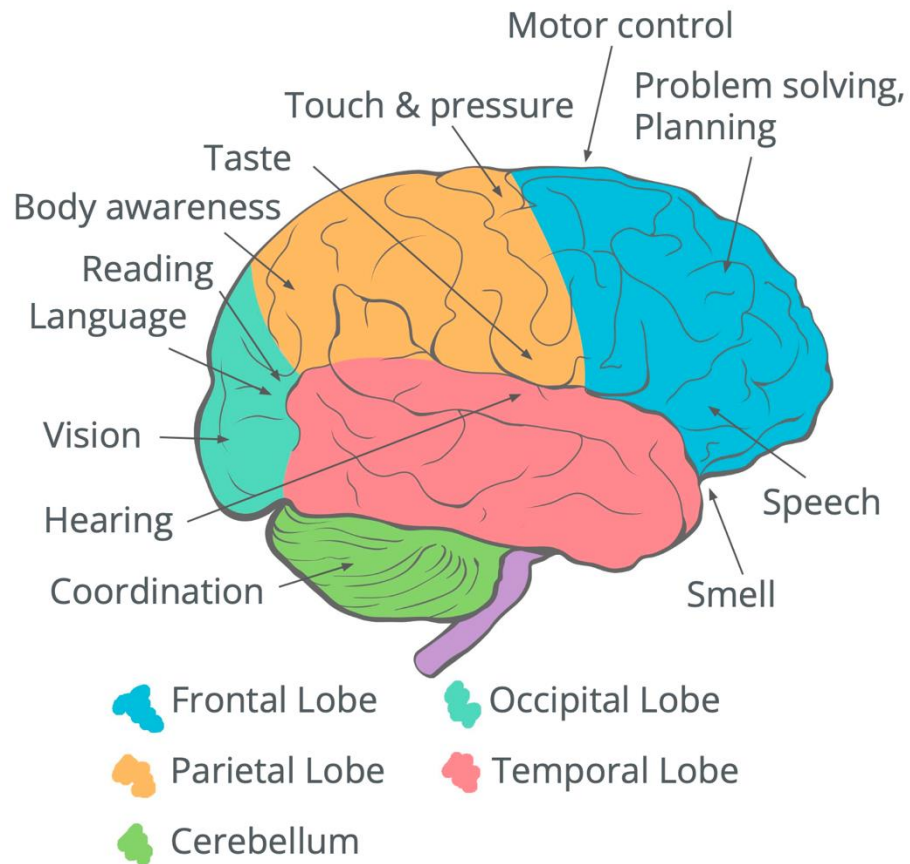
- OpenAI Jukebox - [video](#)
 - **Input:** Artist, Genre, Lyrics
 - **Output:** New music



Architecture of the Brain

Right side/left side (hemispheres)

- Frontal lobe
 - executive functions - memory and planning
- Parietal lobe
 - sensation and spatial awareness
- Temporal lobe (banana shape)
 - hearing and language
- Occipital lobe at back
 - vision from front along optic nerves

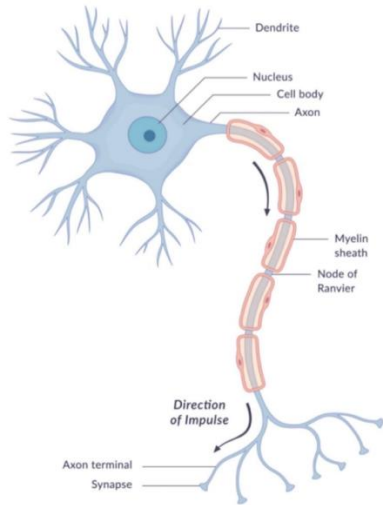


Learning in Nature

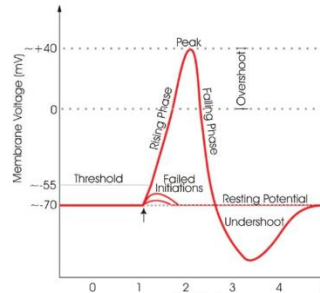
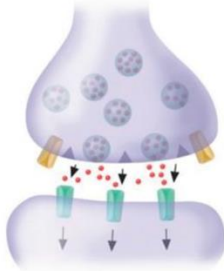
How does the brain work?



1,000s of inputs (other neurons, sensory neurons e.g. taste buds from a salt or sugar molecule...)



- Neurons send out branches called **dendrites**, and a large output called an **axon**
- The axon is coated in myelin that helps it conduct electrical impulses
- The places where the nerve cells make their connections with each other is called a **synapse**



- “Synapse” (from Greek, meaning “to clasp together”)
- **Signals** get **summed up**, and travel to the hillock (Axon neck)
 - If large enough, triggers an action potential travels down axon

1,000s of output targets (e.g. other neurons, muscle cells, gland cells, blood vessels to release hormones...)

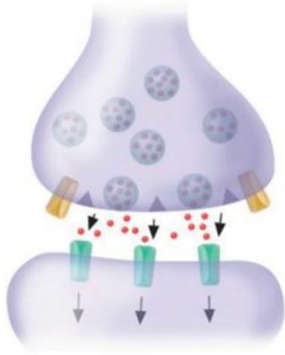
Figure by wetcake (left) and Andrej Kral (right)

Learning in Nature

How does the brain work?



Presynaptic neuron



Postsynaptic neuron

With frequent repeated stimulation, the same level of presynaptic stimulation converts into **greater** postsynaptic potential.

- In other words, *as a neuron gets a lot of practice sending signals to a specific target neuron, it gets better at sending those signals* (the synapse strength increases)
 - Increased strength that lasts for a long time (from minutes to many months) is called **Long Term Potentiation** (weakening is **Long Term Depression**)
- As synapses are strengthened and retain strength, we are able to more easily recall previous experiences.

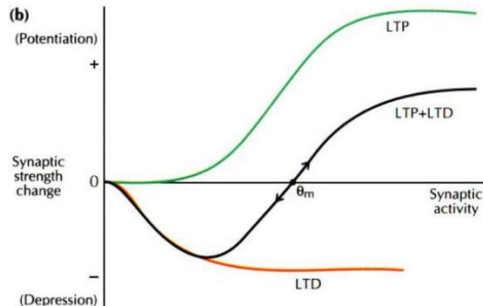
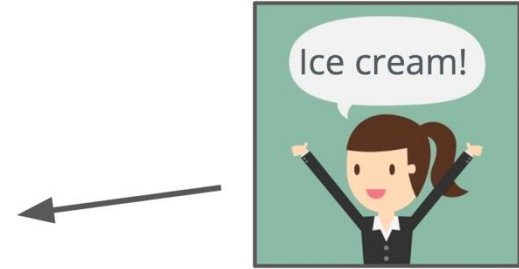
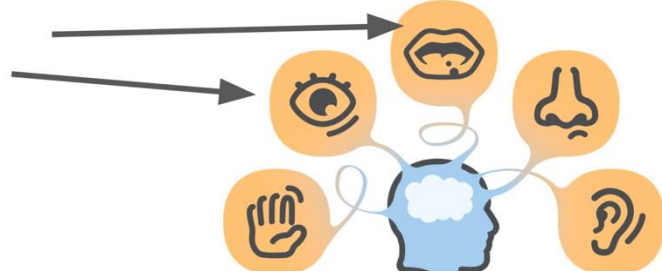


Figure from: "Synaptic Plasticity: A molecular mechanism for metaplasticity", Journal of Current Biology.

Learning in Nature

Hebbian Theory



- **Hebbian theory**

- If two neurons fire at the same time, the connections between them are strengthened, and thus are more likely to fire again together in the future
- If two neurons fire in an uncoordinated manner, their connections are weakened, and they're more likely to act independently in the future

- **Updated Hebbian hypothesis based on recent findings**

- If the presynaptic neuron fires within a window of 20ms before the postsynaptic neuron, the synapse will be strengthened.
- However, if within a window of 20ms after, the synapse will be weakened.

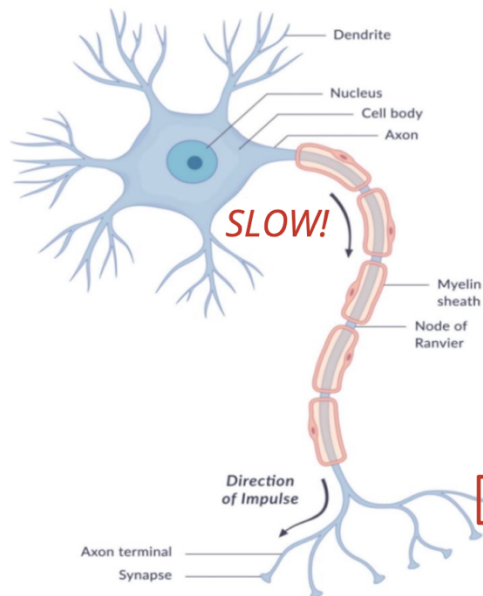
A Brief History

Artificial Neurons



Pitts and McCulloch, 1943

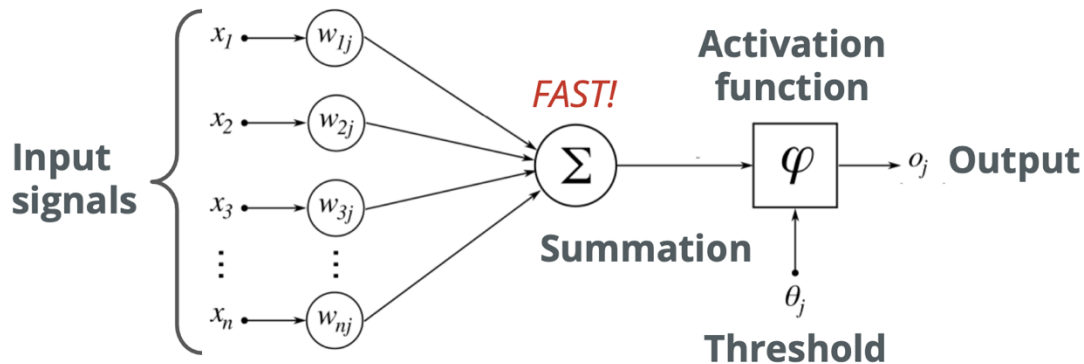
The Artificial Neuron



SLOW!

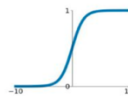
Quantum effects?
([article link](#))

Synaptic weights



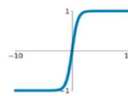
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



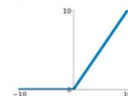
tanh

$$\tanh(x)$$



ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

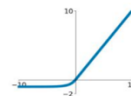


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



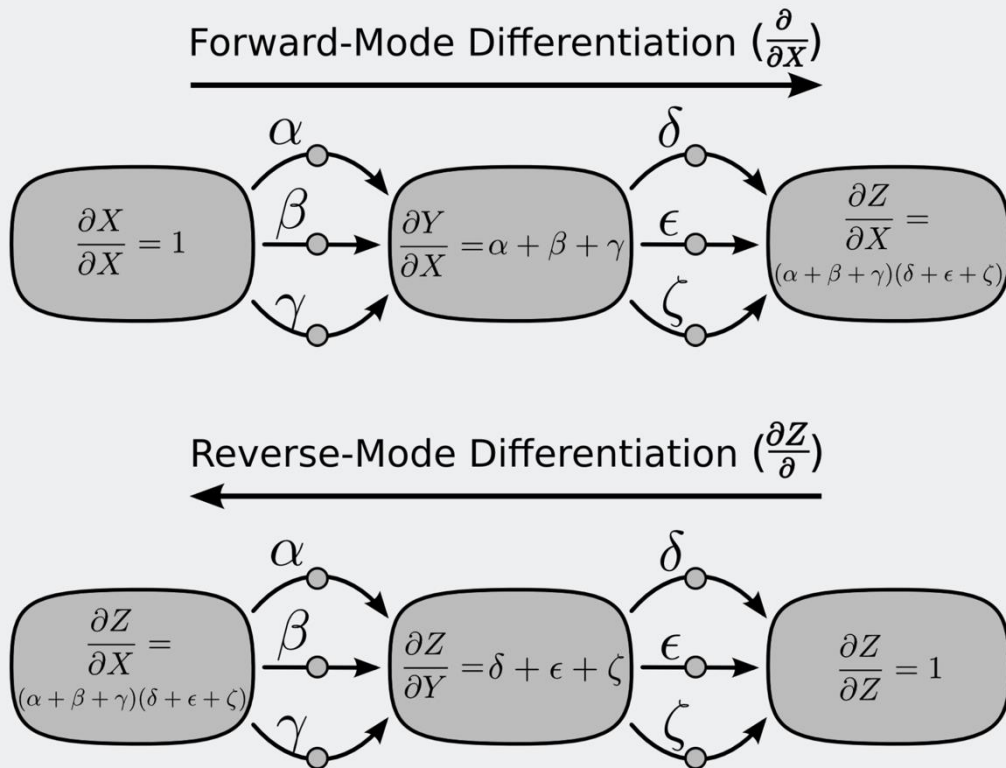


Backpropagation, 1970

The backpropagation algorithm (also called the reverse mode of automatic differentiation) was independently discovered by different researchers.

Seppo Linnainmaa, 1970
Werbos, 1974 (with neural networks)

Reverse-mode differentiation





Rumelhart and Hinton, 1986

Learning representations by back-propagating errors

Nature, 1986. David E. Rumelhart, Geoffrey E. Hinton & Ronald J. Williams

- Backpropagation
- Multiple hidden layers
- Recurrent networks

The second AI winter 1987–1993





History of CNNs

Convolutional neural networks were first introduced by Kuniyiko Fukushima in 1980:

- **1989** - Yann Lecun et al., trained a CNN with “Backpropagation Applied to Handwritten Zip Code Recognition”
- **1998** - Yann Lecun et al., released LeNet5 “Gradient-based learning applied to document recognition”

Example: LeNet5

```
class LeNet(nn.Module):
    def __init__(self):
        super(LeNet, self).__init__()
        self.conv1 = nn.Conv2d(1, 6, 5, padding=2)
        self.conv2 = nn.Conv2d(6, 16, 5)
        self.fc1 = nn.Linear(16*5*5, 120)
        self.fc2 = nn.Linear(120, 84)
        self.fc3 = nn.Linear(84, 10)

    def forward(self, x):
        x = F.max_pool2d(F.relu(self.conv1(x)), (2, 2))
        x = F.max_pool2d(F.relu(self.conv2(x)), (2, 2))
        x = flatten(x)
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = self.fc3(x)
        return x
```




The Current AI Spring



History of NNs on GPUs

- **2004** - first GPU implementation of a neural network.
- **2006** - first GPU implementation of a CNN (just 4 times faster).
- **2012** - AlexNet - won state-of-the-art by significant margin with 60 million parameters.
- **2015** - ImageNet state of the art by a residual network with over 100 layers!



Cloud Computing - GANs & GPT-4

Fake faces generated by StyleGAN2



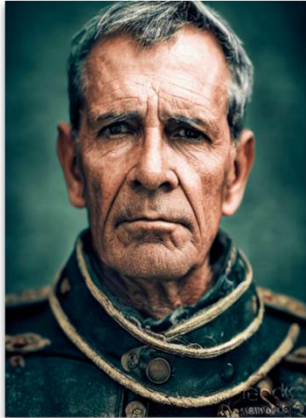
Unsupervised Generative Models

Progressively larger models are being trained on GPU cloud services:

- **2014** - Ian Goodfellow releases “Generative Adversarial Nets”
- **2017** - GoogleBrain releases “Attention is all you need”
- **2019** - OpenAI GPT-2 “Language Models are Unsupervised Multitask Learners”
- **2020** - OpenAI GPT-3 “Language Models are Few-Shot Learners”
- **2022** - The rise of LLMs

2024

Multimodality, Stable Diffusion, Plugins...



"Cyberpunk style image of a motorcycle reflection in rain, ukiyo-e, unreal engine, trending on artstation"

Key Concepts

Overlap with Other Areas of Machine Learning



Machine Learning

Shallow Learning
often hand-engineered

Reinforcement Learning

bad gradients
dynamic not IID
optimising future reward

Deep Learning

multiple layers
gradients
mostly IID

Unsupervised Learning

Generative Models
learning the data
distribution

Meta Learning

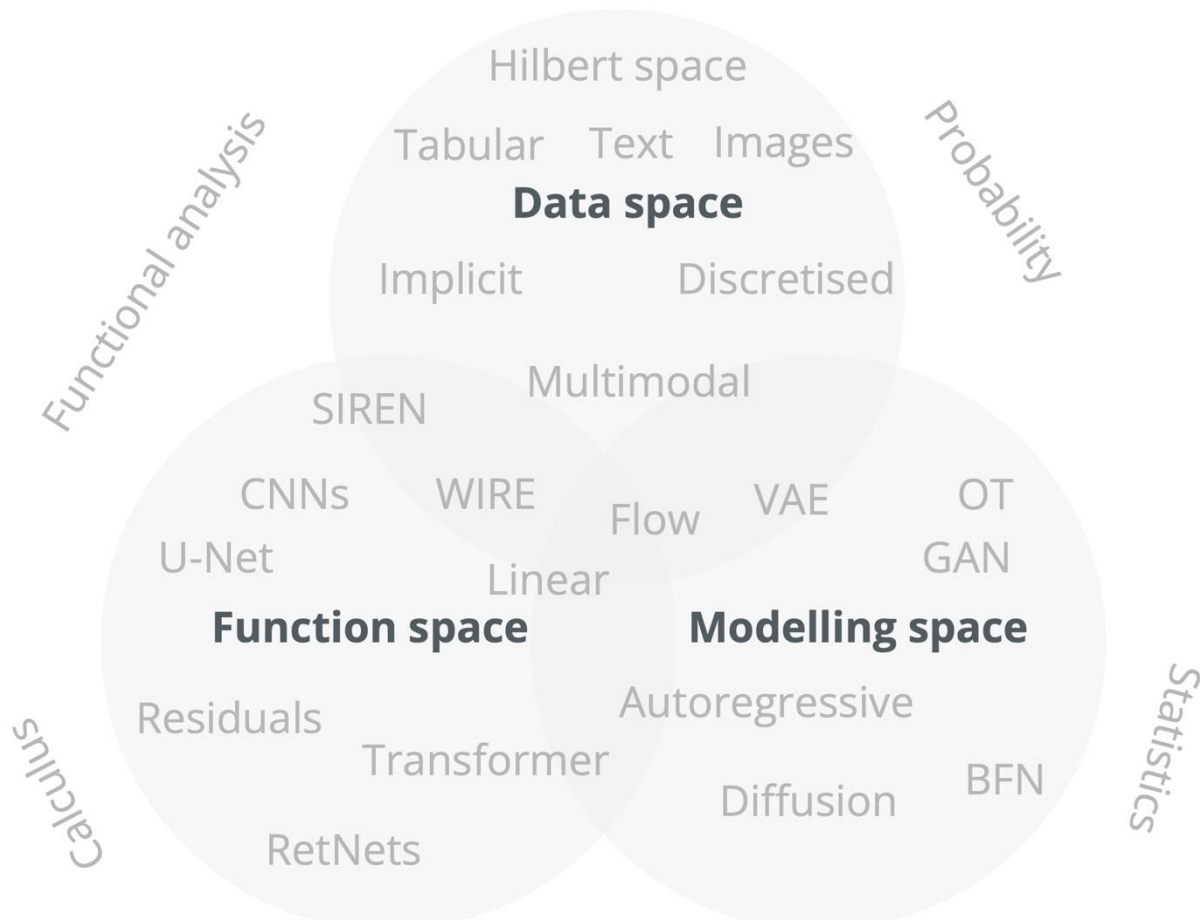
learning the task
distribution

Supervised Learning
learning one task

Discriminative Models
classifying data
regression

Key Concepts

Deep Learning Contributions from Three Spaces



What we learned today!



Summary

In summary, deep learning:

- has overlap with many areas
- achieves state-of-the-art in ill-defined tasks and scales with...
 - very high-dimensional datasets
 - huge datasets
 - parallelisable hardware
- fields of maths: probability, calculus, statistics, functional analysis...
- is rapidly growing and evolving